Module-2

Business Research Design:

Meaning, types and significance of research design, errors affecting research design.

Exploratory Research: Meaning, purpose, methods, Literature search, experience survey, focus groups and comprehensive case methods.

Conclusive Research Design: Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.

Experimental Research Design: Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

Business Research Design

Meaning

The formidable problem that follows the task of defining the research problem is the preparation of the design of the research project, popularly known as the "research design".

Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitutea research design. "A research design is the arrangement of conditions for collection and analysis ofdata in a manner that aims to combine relevance to the research purpose with economy in procedure."[1] In fact, the research design is the conceptual structure within which research is conducted; it constitutesthe blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data. More explicitly, the designing decisions happen to be in respect of:

(i) What is the study about?
(ii) Why is the study being made?
(iii) Where will the study be carried out?
(iv) What type of data is required?
(v) Where can the required data be found?
(vi) What periods of time will the study include?
(vii) What will be the sample design?
(viii) What techniques of data collection will be used?
(ix) How will the data be analysed?
(x) In what style will the report be prepared?

Keeping in view the above stated design decisions, one may split the overall research design intothe following parts:

(a) the sampling design which deals with the method of selecting items to be observed for thegiven study;

(b) the observational design which relates to the conditions under which the observationsare to be made;

(c) the statistical design which concerns with the question of how many items are to beobserved and how the information and data gathered are to be analysed; and

(d) the operational design which deals with the techniques by which the procedures specifiedin the sampling, statistical and observational designs can be carried out.

From what has been stated above, we can state the important features of a research design as under:

(i) It is a plan that specifies the sources and types of information relevant to the researchproblem.

(ii) It is a strategy specifying which approach will be used for gathering and analysing the data.

(iii) It also includes the time and cost budgets since most studies are done under these twoconstraints.

In brief, research design must, at least, contain—(a) a clear statement of the research problem; (b) procedures and techniques to be used for gathering information; (c) the population to be studied;and (d) methods to be used in processing and analysing data.

Types of Research Design

There are different types of research design depend on the nature of the problem and objectives of the study. Following are the four types of research design.

1. Explanatory Research Design
2. Descriptive Research Design
3. Diagnostic Research Design
4. Experimental Research Design

1. Explanatory Research Design

In explanatory research design a researcher uses his own imaginations and ideas. It is based on the researcher personal judgment and obtaining information about something. He is looking for the unexplored situation and brings it to the eyes of the people. In this type of research there is no need of hypothesis formulation.

2. Descriptive Research Design

In descriptive research design a researcher is interested in describing a particular situation or phenomena under his study. It is a theoretical type of researcher design based on the collection designing and presentation of the collected data. Descriptive research design covers the characteristics of people, materials, Scio-economics characteristics such as their age, education, marital status and income etc. The qualitative nature data is mostly collected like knowledge, attitude, beliefs and opinion of the people. Examples of such designs are the newspaper articles, films, dramas, and documentary etc.

3. Diagnostic Research Design

Here researcher wants to know about the root causes of the problem. He describes the factors responsible for the problematic situation. It is a problem solving research design that consists mainly:

1. Emergence of the problem
2. Diagnosis of the problem
3. Solution for the problem and
4. Suggestion for the problem solution

4. Experimental Research Design

In this type of research design is often uses in natural science but it is different in social sciences. Human behavior cannot be measured through test-tubes and microscopes. The social researcher use a method of experiment in that type of research design. One group is subjected to experiment called independent variables while other is considered as control group called dependent variable. The result obtained by the comparison of both the two groups. Both have the cause and effect relationship between each other.

Significance of Research Design

Research design offers the investigator an opportunity to carry out different research operations efficiently. This makes research as valuable as possible producing maximum information with minimum effort, time and money.

Researcher needs to consider all necessary precautions when preparing the design, as any error may upset the whole project. The reliability of result, which a researcher is looking, is proportional with design that constitutes a firm foundation of entire body of research work.

Significance of Research Design in Research Methodology

Research design is significant simply because it allows for the smooth sailing of the various research operations, thus making research as efficient as possible producing maximum information with nominal expenses of effort, time and money.

Just as for better, economical and attractive construction of a home, we require a blueprint (or what is typically known as the map of the home) well planned and prepared by an expert architect, in the same way we require a design or a plan in advance of data collection and analysis for our research study. It means advance planning of the techniques to be implemented for accumulating the appropriate data and the strategies to be employed in their analysis, keeping in view the purpose of the research and the availability of staff, time and money.

Preparation of the design must be carried out meticulously as any error in it may upset the complete project. Research design, actually, has a great significance and impact on the reliability of the results achieved and as such constitutes the firm base of the entire edifice of the research work.

Even then the necessity for a well planned design is at times not realized by many people. The significance which this problem warrants is not given to it. Because of this many researches do not serve the purpose for which they are undertaken. The truth is, they may even provide misleading conclusions.

Thoughtlessness in developing the research project may lead to rendering the research exercise futile. It is, for that reason, crucial that an efficient and appropriate design should be prepared before beginning research operations.

The design assists the researcher to organize his ideas in a form whereby it will be possible for him to watch out for flaws and inadequacies. This type of design can also be given to others for their comments and critical evaluation. In the absence of such a strategy, it will likely be challenging for the critic to supply a comprehensive review of the offered study.

A research design isn’t just a work plan. A work plan details what needs to be done to complete the project but the work plan will flow from the project’s design. The **function of a research design** is to make certain that evidence obtained allows us to answer the initial question as unambiguously as possible. Acquiring relevant evidence involves specifying the kind of evidence required to answer the research question, to test a theory, to judge a programme or to precisely describe some phenomenon.

Errors Affecting Research Designs

The research design must attempt to reduce the 8 types of potential errors that can influence research results, viz.

1.Surrogate Information error
2. Measurement error.
3. Experimental error.
4. Population Specification error.
5. Frame Error.
6. Sampling error.
7. Selection error.
8. Non Response error.

However, it should also be kept in mind that research design must reduce total error, not just one or two aspects of total error.

1. Surrogate Information Error:

Surrogate information error is caused by a variation between the information required to solve the problem and the information sought by the researcher. The so-called price-quality relationship, where a consumer uses the price of a brand to represent its quality level, is a common example of a measure that is subject to surrogate information error (because price level does not always reflect quality level). It has been argued that, in part, the earlier mentioned taste test research done by Coca-Cola India resulted in surrogate information. The company based its decision on taste preferences. The resultant consumer backlash was caused by surrogate information error, as consumers purchase Coke for reasons other than taste alone.

2. Measurement error:

Measurement error is caused by a difference between the information desired by the researcher and the information provided by the measurement process, In other words, not only is it possible to seek the wrong type of information (surrogate information error) but it is also possible to gather information that is different from what is being sought. This is one of the most common and serious errors. For example, respondents may exaggerate their income in order to impress an interviewer; the reported income will then reflect an unknown amount of measurement error, measurement error is particularly difficult to control because it can arise from many different sources.

3. Experimental Error:

Experiments are designed to measure the impact of one or more independent variables on a dependent variable. Experimental error occurs when the effect of the experimental situation itself is measured rather than the effect of the independent variable. For example, a retail chain may increase the price of selected items in four outlets and leave the price of the same items constant in four similar outlets, in an attempt to discover the best pricing strategy. However, unique weather patterns, traffic condition, or competitors' activities may affect the sales at one set of stores and not the other. Thus, the experimental results will reflect the impact of variables other than price. Like measurement error, experimental error can arise from a number of sources.

4. Population Specification Error:

Population specification error is caused by selecting an in appropriate universe or population from which to collect data. This is a potentially serious problem in both industrial and consumer research. A firm wishing to learn the criteria that are considered most important in the purchase of certain machine tools might conduct a survey among purchasing agents. Yet, in many firms, the purchasing agents do not determine or necessarily even know the criteria behind brand selections. These decisions may be made by the machine operators, by committee, or by higher Level executives. A study that focuses on the purchasing agent as the person who decides which brands to order may be subject to population specification error.

5. Frame Error:

The sampling frame is the list of population members from which the sample units are selected. An ideal frame identifies each member of the population once and only once. Frame error is caused by using an inaccurate or incomplete sampling frame. For example, using the telephone directory as a sampling frame for the population of a community contains a potential for frame error. Those families who do not have listed numbers, both voluntarily and non-voluntarily, are likely to differ from those with listed numbers in such respects as income, gender and mobility.

6. Sampling Error:

Sampling error is caused by the generation of a non-representative sample by means of a probability sampling method. For example, a random sample of one hundred university students could produce a sample composed of all females (or all seniors or all business majors). Such a sample would not be representative of the overall student body. Yet it could occur using probability sampling techniques.

7. Selection Error:

Selection error occurs when a non-representative sample is obtained by non-probability sampling methods. For example, if an interviewer was afraid of dogs, it would inevitably happen that in those surveys that allowed any freedom of choice, this interviewer would have avoided homes with dogs present. Obviously, such a practice may introduce error into the survey results. Selection error is a major problem in non-probability samples.

8. Non-response Error:

Non-response error is caused by i) A failure to contact all members of a sample and/or ii)The failure of some contacted members of the sample to respond to all or specific parts of the measurement instrument. Individuals who are difficult to contact or who are reluctant to co-operate will differ, on at least some characteristics, from those who are relatively easy to contact or who readily co-operate. If these differences include the variable of interest, non-response error has occurred. For example, people are more likely to respond to a survey on a topic that interests them. If a firm were to conduct a mall survey to estimate the Incidence of athlete foot among adults, non-response error would be of major concern. Why? Those most likely to be interested in athlete’s foot, and thus most likely to respond to the survey, are current or recent sufferers of the problem. If the firm were to use the percentage of those responding who report having athlete's foot as an estimate of the total population having athlete's foot, the company would probably greatly over estimate the extent of the problem.

Exploratory research: Definition

Exploratory research is defined as a research used to investigate a problem which is not clearly defined. It is conducted to have a better understanding of the existing research problem, but will not provide conclusive results. For such a research, a researcher starts with a general idea and uses this research as a medium to identify issues, that can be the focus for future research. An important aspect here is that the researcher should be willing to change his/her direction subject to the revelation of new data or insight. Such a research is usually carried out when the problem is at a preliminary stage. It is often referred to as grounded theory approach or interpretive research as it used to answer questions like what, why and how.

Types and methodologies of Exploratory research

While it may sound difficult to research something that has very little information about it, there are several methods which can help a researcher figure out the best research design, data collection methods and choice of subjects. There are two ways in which research can be conducted namely primary and secondary.. Under these two types, there are multiple methods which can used by a researcher. The data gathered from these research can be qualitative or quantitative. Some of the most widely used research designs include the following:

Primary research methods

Primary research is information gathered directly from the subject. It can be through a group of people or even an individual. Such a research can be carried out directly by the researcher himself or can employ a third party to conduct it on their behalf. Primary research is specifically carried out to explore a certain problem which requires an in-depth study.

- **Surveys/polls**: Surveys/polls are used to gather information from a predefined group of respondents. It is one of the most important quantitative method. Various types of surveys or polls can be used to explore opinions, trends, etc. With the advancement in technology, surveys can now be sent online and can be very easy to access. For instance, use of a survey app through tablets, laptops or even mobile phones. This information is also available to the researcher in real time as well. Nowadays, most organizations offer short length surveys and rewards to respondents, in order to achieve higher response rates.

For example: A survey is sent to a given set of audience to understand their opinions about the size of mobile phones when they purchase one. Based on such information organization can dig deeper into the topic and make business related decision.

- **Interviews:** While you may get a lot of information from public sources, but sometimes an in person interview can give in-depth information on the subject being studied. Such a research is a qualitative research method. An interview with a subject matter expert can give you meaningful insights that a generalized public source won't be able to provide. Interviews are carried out in person or on telephone which have open-ended questions to get meaningful information about the topic.

For example: An interview with an employee can give you more insights to find out the degree of job satisfaction, or an interview with a subject matter expert of quantum theory can give you in-depth information on that topic.

- **Focus groups:** Focus group is yet another widely used method in exploratory research. In such a method a group of people is chosen and are allowed to express their insights on the topic that is being studied. Although, it is important to make sure that while choosing the individuals in a focus group they should have a common background and have comparable experiences.

For example: A focus group helps a research identify the opinions of consumers if they were to buy a phone. Such a research can help the researcher understand what the consumer value while buying a phone. It may be screen size, brand value or even the dimensions. Based on which the organization can understand what are consumer buying attitudes, consumer opinions, etc.

- **Observations:** Observation research can be qualitative observation or quantitative observation. Such a research is done to observe a person and draw the finding from their reaction to certain parameters. In such a research, there is no direct interaction with the subject.

For example: An FMCG company wants to know how it's consumer react to the new shape of their product. The researcher observes the customers first reaction and collects the data, which is then used to draw inferences from the collective information.

Secondary research methods

Secondary research is gathering information from previously published primary research. In such a research you gather information from sources likes case studies, magazines, newspapers, books, etc.

- **Online research:** In today's world, this is one of the fastest way to gather information on any topic. A lot of data is readily available on the internet and the researcher can download it whenever he needs it. An important aspect to be noted for such a research is the genuineness and authenticity of the source websites that the researcher is gathering the information from.

For example: A researcher needs to find out what is the percentage of people that prefer a specific brand phone. The researcher just enters the information he needs in a search engine and gets multiple links with related information and statistics.

- **Literature research**: Literature research is one of the most inexpensive method used for discovering a hypothesis. There is tremendous amount of information available in libraries, online sources, or even commercial databases. Sources can include newspapers, magazines, books from library, documents from government agencies, specific topic related articles, literature, Annual reports, published statistics from research organizations and so on.

However, a few things have to be kept in mind while researching from these sources. Government agencies have authentic information but sometimes may come with a nominal cost. Also, research from educational institutions is generally overlooked, but in fact educational institutions carry out more number of research than any other entities.

Furthermore, commercial sources provide information on major topics like political agendas, demographics, financial information, market trends and information, etc.

For example: A company has low sales. It can be easily explored from available statistics and market literature if the problem is market related or organization related or if the topic being studied is regarding financial situation of the country, then research data can be accessed through government documents or commercial sources.

- **Case study research:** Case study research can help a researcher with finding more information through carefully analyzing existing cases which have gone through a similar problem. Such analysis are very important and critical especially in today’s business world. The researcher just needs to make sure he analyses the case carefully in regards to all the variables present in the previous case against his own case. It is very commonly used by business organizations or social sciences sector or even in the health sector.

For example: A particular orthopedic surgeon has the highest success rate for performing knee surgeries. A lot of other hospitals or doctors have taken up this case to understand and benchmark the method in which this surgeon does the procedure to increase their success rate.

Exploratory research: Steps to conduct a research

- **Identify the problem**: A researcher identifies the subject of research and the problem is addressed by carrying out multiple methods to answer the questions.
- **Create the hypothesis**: When the researcher has found out that there are no prior studies and the problem is not precisely resolved, the researcher will create a hypothesis based on the questions obtained while identifying the problem.
- **Further research**: Once the data has been obtained, the researcher will continue his study through descriptive investigation. Qualitative methods are used to further study the subject in detail and find out if the information is true or not.

Characteristics of Exploratory research

- They are not structured studies
- It is usually low cost, interactive and open ended.
- It will enable a researcher answer questions like what is the problem? What is the purpose of the study? And what topics could be studied?
- To carry out exploratory research, generally there is no prior research done or the existing ones do not answer the problem precisely enough.
- It is a time consuming research and it needs patience and has risks associated with it.
- The researcher will have to go through all the information available for the particular study he is doing.
- There are no set of rules to carry out the research per se, as they are flexible, broad and scattered.
- The research needs to have importance or value. If the problem is not important in the industry the research carried out is ineffective.
- The research should also have a few theories which can support its findings as that will make it easier for the researcher to assess it and move ahead in his study
- Such a research usually produces qualitative data, however in certain cases quantitative data can be generalized for a larger sample through use of surveys and experiments.

Advantages of Exploratory research

- The researcher has a lot of flexibility and can adapt to changes as the research progresses.
- It is usually low cost.
- It helps lay the foundation of a research, which can lead to further research.
- It enables the researcher understand at an early stage, if the topic is worth investing the time and resources and if it is worth pursuing.
- It can assist other researchers to find out possible causes for the problem, which can be further studied in detail to find out, which of them is the most likely cause for the problem.

Disadvantages of Exploratory research

- Even though it can point you in the right direction towards what is the answer, it is usually inconclusive.
- The main disadvantage of exploratory research is that they provide qualitative data. Interpretation of such information can be judgmental and biased.
- Most of the times, exploratory research involves a smaller sample, hence the results cannot be accurately interpreted for a generalized population.
- Many a times, if the data is being collected through secondary research, then there is a chance of that data being old and is not updated.

Importance of Exploratory research

Exploratory research is carried out when a topic needs to be understood in depth, especially if it hasn't been done before. The goal of such a research is to explore the problem and around it and not actually derive a conclusion from it. Such kind of research will enable a researcher to set a strong foundation for exploring his ideas, choosing the right research design and finding variables that actually are important for the analysis. Most importantly, such a research can help organizations or researchers save up a lot of time and resources, as it will enable the researcher to know if it worth pursuing.

Conclusive Research Design

Conclusive research design, as the name implies, is applied to generate findings that are practically useful in reaching conclusions or decision-making. In this type of studies research objectives and data requirements need to be clearly defined. Findings of conclusive studies usually have specific uses. Conclusive research design provides a way to verify and quantify findings of exploratory studies.

Conclusive research design usually involves the application of quantitative methods of data collection and data analysis. Moreover, conclusive studies tend to be deductive in nature and research objectives in these types of studies are achieved via testing hypotheses.

Factor	**Conclusive**	**Exploratory**
Objectives	To test hypothesis and relationships	To get insights and understanding
Characteristics	Information needs a clearly defined Research process is formal and structured Large representative sample Data analysis is quantitative	Information needs are loosely defined Research process is unstructured and flexible Small, non-representative sample Primary data analysis is qualitative
Findings	Conclusive	Only tentative
Outcome	Findings used as input to decision making	Generally followed by further exploratory conclusive research

The table below illustrates the main differences between conclusive and exploratory research design:

Main differences between conclusive and exploratory research design

It has to be noted that "conclusive research is more likely to use statistical tests, advanced analytical techniques, and larger sample sizes, compared with exploratory studies. Conclusive research is more likely to use quantitative, rather than qualitative techniques"[1]. Conclusive research is helpful in providing a reliable or representative picture of the population through the application of valid research instrument.

Conclusive research design can be divided into two categories: descriptive research and causal research.

Descriptive research is used to describe some functions or characteristics of phenomenon and can be further divided into the following groups:

1. Case study;
2. Case series study;
3. Cross-sectional study;
4. Longitudinal study;
5. Retrospective study.

Causal research, on the other hand, is used to research cause and affect relationships. Two popular research methods for causal studies are experimental and quasi-experimental studies.

Descriptive Research

Meaning

The descriptive research design involves observing and collecting data on a given topic without attempting to infer cause-and-effect relationships. The goal of descriptive research is to provide a comprehensive and accurate picture of the population or phenomenon being studied and to describe the relationships, patterns, and trends that exist within the data.

Descriptive research methods can include surveys, observational studies, and case studies, and the data collected can be qualitative or quantitative. The findings from descriptive research provide valuable insights and inform future research, but do not establish cause-and-effect relationships.

Types of Descriptive Research

1. Survey Research

Surveys are a type of descriptive research that involves collecting data through self-administered or interviewer-administered questionnaires. Additionally, they can be administered in-person, by mail, or online, and can collect both qualitative and quantitative data.

2. Observational Research

Observational research involves observing and collecting data on a particular population or phenomenon without manipulating variables or controlling conditions. It can be conducted in naturalistic settings or controlled laboratory settings.

3. Case Study Research

Case study research is a type of descriptive research that focuses on a single individual, group, or event. It involves collecting detailed information on the subject through a variety of methods, including interviews, observations, and examination of documents.

4. Focus Group Research

Focus group research involves bringing together a small group of people to discuss a particular topic or product. Furthermore, the group is usually moderated by a researcher and the discussion is recorded for later analysis.

5. Ethnographic Research

Ethnographic research involves conducting detailed observations of a particular culture or community. It is often used to gain a deep understanding of the beliefs, behaviors, and practices of a particular group.

Cross sectional studies and longitudinal studies.

Cross-sectional study

Both the cross-sectional and the longitudinal studies are observational studies. This means that researchers record information about their subjects without manipulating the study environment. In our study, we would simply measure the cholesterol levels of daily walkers and non-walkers along with any other characteristics that might be of interest to us. We would not influence non-walkers to take up that activity, or advise daily walkers to modify their behaviour. In short, we'd try not to interfere.

The defining feature of a cross-sectional study is that it can compare different population groups at a single point in time. Think of it in terms of taking a snapshot. Findings are drawn from whatever fits into the frame.

To return to our example, we might choose to measure cholesterol levels in daily walkers across two age groups, over 40 and under 40, and compare these to cholesterol levels among non-walkers in the same age groups. We might even create subgroups for gender. However, we would not consider past or future cholesterol levels, for these would fall outside the frame. We would look only at cholesterol levels at one point in time.

The benefit of a cross-sectional study design is that it allows researchers to compare many different variables at the same time. We could, for example, look at age, gender, income and educational level in relation to walking and cholesterol levels, with little or no additional cost.

However, cross-sectional studies may not provide definite information about cause-and-effect relationships. This is because such studies offer a snapshot of a single moment in time; they do not consider what happens before or after the snapshot is taken. Therefore, we can't know for sure if our daily walkers had low cholesterol levels before taking up their exercise regimes, or if the behaviour of daily walking helped to reduce cholesterol levels that previously were high.

Longitudinal Study

A longitudinal study, like a cross-sectional one, is observational. So, once again, researchers do not interfere with their subjects. However, in a longitudinal study, researchers conduct several observations of the same subjects over a period of time, sometimes lasting many years.

The benefit of a longitudinal study is that researchers are able to detect developments or changes in the characteristics of the target population at both the group and the individual level. The key here is that longitudinal studies extend beyond a single moment in time. As a result, they can establish sequences of events.

To return to our example, we might choose to look at the change in cholesterol levels among women over 40 who walk daily for a period of 20 years. The longitudinal study design would account for cholesterol levels at the onset of a walking regime and as the walking behaviour continued over time. Therefore, a longitudinal study is more likely to suggest cause-and-effect relationships than a cross-sectional study by virtue of its scope.

In general, the research should drive the design. But sometimes, the progression of the research helps determine which design is most appropriate. Cross-sectional studies can be done more quickly than longitudinal studies. That's why researchers might start with a cross-sectional study to first establish whether there are links or associations between certain variables. Then they would set up a longitudinal study to study cause and effect.

Experimental Design Definition

In Statistics, the experimental design or the design of experiment (DOE) is defined as the design of an information-gathering experiment in which a variation is present or not, and it should be performed under the full control of the researcher. This term is generally used for controlled experiments. These experiments minimise the effects of the variable to increase the reliability of the results. In this design, the process of an experimental unit may include a group of people, plants, animals, etc.

Classification/Types of Experimental Designs

There are different types of experimental designs of research. They are:

1. Formal & Informal Research Design
2. Pre-experimental Research Design
3. True-experimental Research Design
4. Quasi-Experimental Research Design
5. Statistical experimental design.
6. Randomised Block Design

In this article, we are going to discuss these different experimental designs for research with examples.

Formal Research

Formal research is a type of research study conducted using a systematic approach and scientific methods. A formal research study typically involves several components: abstract, introduction, literature review, research design and method, results and analysis, conclusion, bibliography. All in all, formal research is more scientific in nature than informal research and aims to discover new information or solve a problem. It always uses up to date, reliable sources. Moreover, the sources of information used in research are also complied using a reference system.
Researchers usually begin a formal research study with a hypothesis; then, they test this hypothesis rigorously. They also explore and analyze the literature already available on their research subject. This allows them to study the research subject from multiple perspectives, acknowledging different problems that need to be solved. In comparison to informal research, formal research tends to be objective or unbiased. This mainly happens because formal research uses scientific methods which involve objective information. If the research involves participants, for example, in surveys or interviews, the researcher has to obtain their written consent first. Furthermore, we can apply the findings from formal research to a larger audience.

Informal Research

Informal research is the use of nonscientific methods to gather and analyze data. Since the research uses nonscientific methods, its finding cannot be applied to a wider audience. Most of us conduct informal research on a regular basis. For example, conducting a simple survey, informal interviews with a target audience, seeking opinions of colleagues, searching through company files, etc. are all nonscientific methods of gathering data and are therefore forms of informal research.

Informal research does not take much time or effort. This usually involves just looking through some information or gathering some data. For example, suppose you are going to buy a new laptop. You will ask for the opinions of your colleagues and may look through some websites and reviews to know more information about laptop brands. Your decision here will be based on informal research. But it’s important to know this information may be subjective. Moreover, as this is an informal research, you might also not worry about the source of information.

Difference Between Formal and Informal Research

Definition
Formal research is a type of research study conducted using a systematic approach and scientific methods while informal research is the use of nonscientific methods to gather and analyze data.
Methods
While formal research uses scientific methods, informal research uses nonscientific methods
Time and Effort

Informal research takes less time and effort than formal research.

Objectivity

Formal research tends to be more objective or unbiased than informal research.

Sources

It's necessary to list the sources of information in a formal research, but informal research does not follow this practice.

Application of Findings

Findings of formal research can be applied to a larger group, whereas findings of informal research can be applied only to a smaller group.

Pre-experimental Research Design

The simplest form of experimental research design in Statistics is the pre-experimental research design. In this method, a group or various groups are kept under observation, after some factors are recognised for the cause and effect. This method is usually conducted in order to understand whether further investigations are needed for the targeted group. That is why this process is considered to be cost-effective. This method is classified into three types, namely,

- Static Group Comparison
- One-group Pretest-posttest Experimental Research Design
- One-shot Case Study Experimental Research Design

True-experimental Research Design

This is the most accurate form of experimental research design as it relies on the statistical hypothesis to prove or disprove the hypothesis. This is the most commonly used method implemented in Physical Science. True experimental research design is the only method that establishes the cause and effect relationship within the groups. The factors which need to be satisfied in this method are:

- Random variable
- Variable can be manipulated by the researcher
- Control Groups (A group of participants are familiar with the experimental group, but the experimental rules do not apply to them)
- Experimental Group (Research participants where experimental rules are applied)

Quasi-Experimental Design

A quasi-experimental design is similar to a true experimental design, but there is a difference between the two.

In a true experiment design, the participants of the group are randomly assigned. So, every unit has an equal chance of getting into the experimental group.

In a quasi-experimental design, the participants of the groups are not randomly assigned. So, the researcher cannot make a cause or effect conclusion. Thus, it is not possible to assign the participants to the group.

Apart from these types of experimental design research in statistics, there are other two methods used in the research process such as randomized block design and completely randomized design.

Statistical experimental design.

The (statistical) design of experiments (DOE) is an efficient procedure for planning experiments so that the data obtained can be analyzed to yield valid and objective conclusions. DOE begins with determining the objectives of an experiment and selecting the process factors for the study.

4 types of experimental design statistics

Four major design types with relevance to user research are experimental, quasi-experimental, correlational and single subject. These research designs proceed from a level of high validity and generalizability to ones with lower validity and generalizability.
the statistical design process?

The Statistical Process has five steps:

- Design the study
- Collect the data
- Describe the data
- Make inferences
- Take action.

In a designed experiment, researchers control the conditions of the study. In an observational study, researchers don't control the conditions but only observe what happens.

Randomised Block Design

The randomised block design is preferred in the case when the researcher is clear about the distinct difference among the group of objects. In this design, the experimental units are classified into subgroups of similar categories. Those groups are randomly assigned to the group of treatment. The blocks are classified in such a way in the variability within each block should be less than the variability among the blocks. This block design is quite efficient as it reduces the variability and produces a better estimation.

Example:

In a drug testing experiment, the researcher believes that age is the most significant factor. So he divides the units according to the age groups such as

- Under 15 years old
- 15 – 35 years old
- 36 – 55 years old
- Over 55 years old

Completely Randomised Design

Of all the types, the simplest type of experimental design is the completely randomized design, in which the participants are randomly assigned to the treatment groups. The main advantage of using this method is that it avoids bias and controls the role of chance. This method provides a solid foundation for Statistical analysis as it allows the use of probability theory.

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